Gradient of a line

1

a The line has a gradient of zero.

b The line has an undefined gradient.

c The line has a positive gradient.

d The line has a negative gradient.

2

a i 2 ii 6 iii $\frac{1}{3}$

b i 12 ii 6 iii 2

c i 14 ii 7 iii 2

d i 12 ii 4 iii 3

e i -12 ii 4 iii -3

f i -6 ii 2 iii -3

g i -6 ii 6 iii -1

h i 8 ii 8 iii 1

3 Because a vertical line has no run. So the formula for the gradient $\frac{\text{rise}}{\text{run}}$ will have a denominator of zero, and any fraction with a sero denominator is undefined.

4 0

5

a 2

b $-\frac{7}{5}$

c $\frac{1}{2}$

d $-\frac{2}{3}$

Gradient formula (Extended)

6 $-\frac{3}{2}$

7 No

8

a -2

b As x increases, the value of y decreases.

9

a 3

b $\frac{3}{5}$

c -4

d -3

10

a $\frac{5}{3}$

b $\frac{5}{3}$

c Yes

11 $m=1$

12

a $t=3$

b $t=15$

c $d=20$

d $c=-\frac{12}{7}$

13

a $Q\,(4,-2)$

b $R\,(-4,-18)$

c 2

14

a $y=27$

b 5

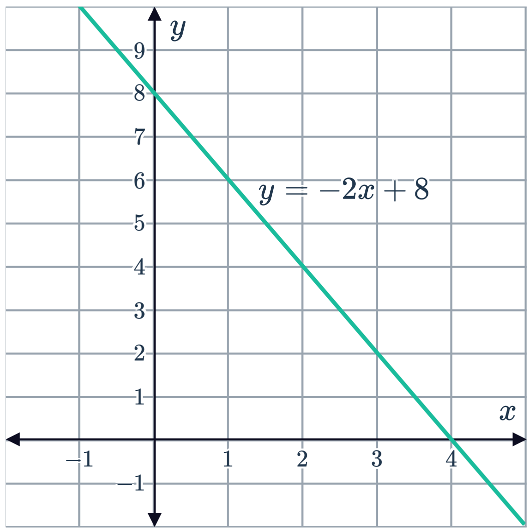
15

a

i $y=8$

ii $x=4$

b



c -2

16

a $y=1$

b $y=1$

c

i 1

ii -1

d -1

Properties of polygons (Extended)

17

a

i $-\frac{2}{3}$

ii $-\frac{2}{3}$

iii 1

iv 1

b A parallelogram because opposite sides are parallel because they have the same gradient.

18

a -1

b 1

c -1

d Yes

19

a -1

b -1

c Yes

d Yes

e Parallelogram

20

a $D\left(\frac{13}{2}, -4\right)$ and $E\left(\frac{1}{2}, -\frac{17}{2}\right)$

b $\frac{3}{4}$

c $\frac{3}{4}$

d Yes

21

a BC

b 15 units^2

Applications

22

a 0.9

b No

23

a -0.72

b -0.35

c Run A

24 890 m